

Problem 2.1

Let $B \subseteq \mathbb{R}^d$.

- (a) Show that if μ is a Borel measure on $\mathbb{R}^d$ then $\mu|_B$ is also a Borel measure on $\mathbb{R}^d$.
- (b) Show that if μ is a Borel regular measure on $\mathbb{R}^d$, if B is μ -measurable and of finite measure then $\nu := \mu|_B$ is a Radon measure on $\mathbb{R}^d$.

Solution.

Problem 2.2*

Suppose μ is an outer measure on X , $E_1, \dots, E_n$ are disjoint subsets of X and $c_1, \dots, c_n$ are distinct nonzero real numbers. Prove that $c_1\chi_{E_1} + \dots + c_n\chi_{E_n}$ is a μ -measurable function if and only if $E_1, \dots, E_n$ are μ -measurable sets.

Solution.

Problem 2.3*

Suppose $X \subseteq \mathbb{R}$ is a Borel set and $f : X \rightarrow \mathbb{R}$ is a function such that $\{x \in X : f \text{ is not continuous at } x\}$ is a countable set. Prove that f is a Borel measurable function.

Solution.

Problem 2.4*

Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at every element of $\mathbb{R}$. Prove that f' is a Borel measurable function from $\mathbb{R}$ to $\mathbb{R}$.

Solution.

Problem 2.5*

Let h be a continuous and strictly increasing real valued function on $\mathbb{R}$. Prove that if $B \subseteq \mathbb{R}$, then $h(B)$ is a Borel set if and only if B is a Borel set.

Solution.

Problem 2.6*

Suppose $f : B \rightarrow \mathbb{R}$ is a Borel measurable function. Define $g : \mathbb{R} \rightarrow \mathbb{R}$ to be the function which coincides with f on B and is identically equal to 0 on $\mathbb{R} \setminus B$. Prove that g is a Borel measurable function.

Solution.

Problem 2.7*

Say whether or not every open subset of $\mathbb{R}^d$ is a countable union of closed subsets of $\mathbb{R}^d$. Justify your answer (your argument should be short, say, half a page).

Solution.

Problem 2.8

Let μ be an outer measure on a nonempty set X . Let $f, g : X \rightarrow [-\infty, +\infty]$ be two μ -summable functions. Assume that $A, B \subseteq X$ are two μ -measurable sets. Recall

$$\int_A f \, d\mu = \int_X \chi_A f \, d\mu.$$

- (a) Prove that f is $\mu|_A$ -summable and $\int_X f \, d(\mu|_A) = \int_X f \cdot \chi_A \, d\mu$.
- (b) Prove that $\int_X (f + g) \, d\mu = \int_X f \, d\mu + \int_X g \, d\mu$, and $\int_X cf \, d\mu = c \int_X f \, d\mu$, for all $c \in \mathbb{R}$.
- (c) Prove that if $f \leq g$ then $\int_X f \, d\mu \leq \int_X g \, d\mu$.

Solution.

Problem 2.9

Let μ be an outer measure on a nonempty set X , let $\{A_k\}_{k=1}^\infty$ be a collection of pairwise disjoint μ -measurable subsets of X , and let $f : X \rightarrow \mathbb{R}$ be a summable function. Prove that

$$\int_A f \, d\mu = \sum_{k=1}^\infty \int_{A_k} f \, d\mu,$$

where $A = \bigcup_{k=1}^\infty A_k$.

Solution.

Problem 2.10*

Let (Ω, μ) be an outer measure space and suppose $f_n, g_n, f, g \in L^1(\mu)$. Suppose

$$|f_n| \leq g_n, \quad f_n \rightarrow f \text{ a.e.}, \quad g_n \rightarrow g \text{ a.e.}, \quad \lim_{n \rightarrow \infty} \int_\Omega g_n(x) \mu(dx) = \int_\Omega g(x) \mu(dx).$$

Show that $\lim_{n \rightarrow \infty} \int_\Omega f_n(x) \mu(dx) = \int_\Omega f(x) \mu(dx)$.

Solution.